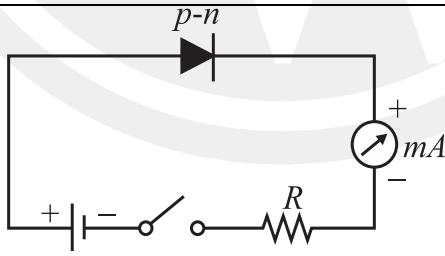
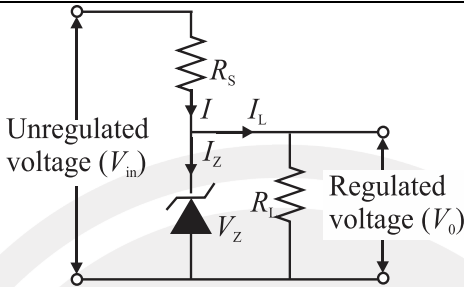
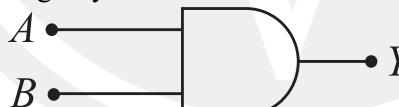
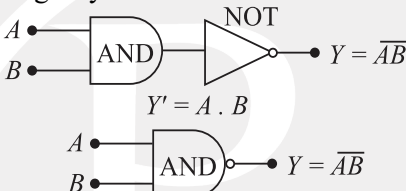
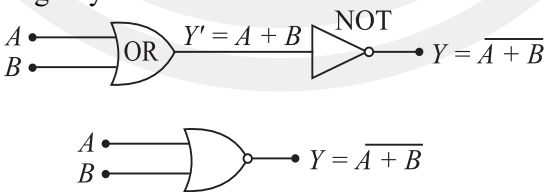


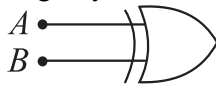
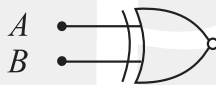
# Semiconductor Electronics

| Sr. No. | Concept                        | Formulas                                                                                                                                                                                                                                                                                                                                                   | Other information                                                                                                                                                                                                                                                                                                                      |
|---------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Energy band gap                | <p>The energy band gap: <math>E_g = E_C - E_V</math></p> <p>If <math>\lambda</math> is the wavelength of radiation used in shifting the electron;</p> $E_g = h\nu = \frac{hc}{\lambda}$                                                                                                                                                                    | <p><math>h</math> = Planck's constant<br/> <math>c</math> = velocity of light<br/> <math>E_V</math> = valence band energy<br/> <math>E_C</math> = conduction band energy<br/> <math>\nu</math> = frequency of radiation</p>                                                                                                            |
| 2.      | Conductivity of semiconductors | <p>Mobility of a charge carrier, <math>\mu = \frac{v}{E}</math></p> <p>Electric current in intrinsic semiconductors<br/> <math>I = eA(n_e v_e + n_h v_h)</math>.</p> <p>Electrical resistivity, <math>\rho = \frac{1}{e(n_e \mu_e + n_h \mu_h)}</math></p> <p>Electrical conductivity, <math>\sigma = \frac{1}{\rho} = e(n_e \mu_e + n_h \mu_h)</math></p> | <p><math>\rho</math> = electrical resistivity<br/> <math>E</math> = electric field<br/> <math>v</math> = velocity of electron<br/> <math>n_e, n_h</math> = concentration of electrons and holes respectively.<br/> <math>\mu_e, \mu_h</math> = electron and hole mobility respectively<br/> <math>A</math> = area of cross-section</p> |
| 3.      | p-n junction diode             |  <p>The d.c. resistance of a junction diode, <math>r_{dc} = \frac{V_d}{I}</math></p> <p>The dynamic or a.c. resistance of a junction diode,<br/> <math>r_{ac} = \frac{\Delta V_d}{\Delta I}</math></p> <p>Applied voltage, <math>V = V_d + IR</math></p>                | <p><math>I</math> = current flowing through the diode<br/> <math>R</math> = external resistance<br/> <math>\Delta V_d</math> = change in voltage in diode<br/> <math>\Delta I</math> = change in current in diode<br/> <math>V_d</math> = voltage across diode.</p>                                                                    |

| Sr. No. | Concept                               | Formulas                                                                                                                                                                                                                                        | Other information                                                                                                                              |
|---------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.      | Mass action law                       | $n_e n_h = n_i^2$<br>In n type semiconductors;<br>$n_e \gg n_h$<br>In p type semiconductors:<br>$n_h \gg n_e$                                                                                                                                   | $n_i$ = intrinsic carrier concentration<br>$n_h$ = concentration of holes<br>$n_e$ = concentration of electrons                                |
| 5.      | Zener diode                           |  <p>For a Zener diode, <math>V_Z = V_{in} - R_s I</math></p> <p>Series resistor, <math>R_s = \frac{V_{in} - V_Z}{I}</math></p>                                 | $I_Z$ = current through Zener diode.<br>$R_L$ = load resistance<br>$V_0$ = output voltage<br>$V_{in}$ = input voltage<br>$V_Z$ = Zener voltage |
| 6.      | Half wave and full wave rectifier     | In half wave rectifier;<br>Output ripple frequency = input ripple frequency<br>In full wave rectifier;<br>Output ripple frequency = $2 \times$ input ripple frequency                                                                           |                                                                                                                                                |
| 7.      | Sum and differences of binary numbers | The sum of digits 0 and 1:<br>(i) $0 + 0 = 0$ (ii) $1 + 0 = 1$<br>(iii) $0 + 1 = 1$ (iv) $1 + 1 = 10$<br>The four laws of difference (in binary system) are as below<br>(i) $0 - 0 = 0$ (ii) $1 - 0 = 1$<br>(iii) $1 - 1 = 0$ (iv) $10 - 1 = 1$ |                                                                                                                                                |
| 8.      | Boolean expression                    | (i) Boolean postulates<br>(a) $0 + A = A$ (b) $1 + A = 1$<br>(c) $0 \cdot A = 0$ (d) $1 \cdot A = A$<br>(e) $A + \bar{A} = 1$<br>(ii) Identity law,<br>(a) $A + A = A$ (b) $A \cdot A = A$<br>(iii) Negation law $\bar{\bar{A}} = A$            | $A$ = logic input<br>0 = low state<br>1 = high state                                                                                           |
| 9.      | De-Morgan's theorem                   | (a) $\overline{A+B} = \bar{A} \cdot \bar{B}$ (b) $\overline{A \cdot B} = \bar{A} + \bar{B}$                                                                                                                                                     | $A, B$ = logic inputs                                                                                                                          |

| Sr. No. | Concept                 | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Other information                                                                             |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
|---------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--|--------|-----|-----|-----------------|---|---|---|---|---|---|---|---|---|---|---|---|------------------------------------------------------|
| 10.     | Laws of Boolean algebra | Commutative laws<br>(a) $A + B = B + A$ (b) $A \cdot B = B \cdot A$<br>Associative laws,<br>(a) $A + (B + C) = (A + B) + C$<br>(b) $A \cdot (B \cdot C) = (A \cdot B) \cdot C$<br>Distributive laws<br>(a) $A \cdot (B + C) = A \cdot B + A \cdot C$<br>(b) $(A + B) \cdot (A + C) = A + BC$<br>Boolean identities<br>(a) $A + \overline{A} B = A + B$<br>(b) $A ( \overline{A} + B ) = AB$<br>(c) $A + BC = (A + B)(A + C)$                                                                                | $A, B, C$ = logic inputs<br>$\overline{A}, \overline{B}$ = complements of logic input A and B |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 11.     | OR gate                 | Boolean expression:<br><div><math>Y = A + B</math></div> Logic symbol:<br><br>Truth table<br><table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th><math>A</math></th><th><math>B</math></th><th><math>Y = A + B</math></th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>           | Input                                                                                         |  | Output | $A$ | $B$ | $Y = A + B$     | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | $Y$ = logic output<br>$A, B$ = two inputs of OR gate |
| Input   |                         | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| $A$     | $B$                     | $Y = A + B$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 0       | 0                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 0       | 1                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 1       | 0                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 1       | 1                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 12.     | AND gate                | Boolean expression:<br><div><math>Y = A \cdot B</math></div> Logic symbol:<br><br>Truth table<br><table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th><math>A</math></th><th><math>B</math></th><th><math>Y = A \cdot B</math></th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> | Input                                                                                         |  | Output | $A$ | $B$ | $Y = A \cdot B$ | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |                                                      |
| Input   |                         | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| $A$     | $B$                     | $Y = A \cdot B$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 0       | 0                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 0       | 1                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 1       | 0                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |
| 1       | 1                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                               |  |        |     |     |                 |   |   |   |   |   |   |   |   |   |   |   |   |                                                      |

| Sr. No. | Concept            | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Other information          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|---------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------|--------|--------------------|-----|-----|------------------|----------------------------|------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| 13.     | NOT gate           | <p>Boolean expression:</p> $Y = \overline{A}$ <p>Logic symbol:</p>  <p>Truth table</p> <table><tr><th>Input</th><th>Output</th></tr><tr><td><math>A</math></td><td><math>Y = \overline{A}</math></td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>                                                                                                                                                                                                                                                                                        | Input                      | Output | $A$    | $Y = \overline{A}$ | 0   | 1   | 1                | 0                          | $A$ = input of NOT gate<br>$Y$ = logic output. |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| Input   | Output             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| $A$     | $Y = \overline{A}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 0       | 1                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 1       | 0                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                            |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 14.     | NAND gate          | <p>Boolean expression:</p> $Y = \overline{A \cdot B} \text{ or } Y = \overline{A} \overline{B}$ <p>Logic symbol:</p>  <p>Truth table</p> <table><tr><th colspan="2">Input</th><th colspan="2">Output</th></tr><tr><th><math>A</math></th><th><math>B</math></th><th><math>Y' = A \cdot B</math></th><th><math>Y = \overline{A \cdot B}</math></th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table> | Input                      |        | Output |                    | $A$ | $B$ | $Y' = A \cdot B$ | $Y = \overline{A \cdot B}$ | 0                                              | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 0 |  |
| Input   |                    | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| $A$     | $B$                | $Y' = A \cdot B$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $Y = \overline{A \cdot B}$ |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 0       | 0                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 0       | 1                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 1       | 0                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 1       | 1                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 15.     | NOR gate           | <p>Boolean expression:</p> $Y = \overline{A + B}$ <p>Logic symbol:</p>  <p>Truth table</p> <table><tr><th colspan="2">Input</th><th colspan="2">Output</th></tr><tr><th><math>A</math></th><th><math>B</math></th><th><math>Y' = A + B</math></th><th><math>Y = \overline{A + B}</math></th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>                                                    | Input                      |        | Output |                    | $A$ | $B$ | $Y' = A + B$     | $Y = \overline{A + B}$     | 0                                              | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 |  |
| Input   |                    | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| $A$     | $B$                | $Y' = A + B$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $Y = \overline{A + B}$     |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 0       | 0                  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 0       | 1                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 1       | 0                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 1       | 1                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0                          |        |        |                    |     |     |                  |                            |                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |

| Sr. No. | Concept    | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Other information |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
|---------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------------------------------------------------------------------------------------|
| 16.     | X-OR gate  | <p>Boolean expression:<br/><math>Y = A \oplus B</math> or <math>Y = \overline{A}B + A\overline{B}</math></p> <p>Logic symbol:</p>  $Y = A \cdot \overline{B} + B \cdot \overline{A}$ <table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>                                                                                                                          | Input             |  | Output | A | B | Y | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | <p><math>Y</math> = output value<br/><math>A, B</math> =input value of the gate.</p> |
| Input   |            | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| A       | B          | Y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0       | 0          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0       | 1          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1       | 0          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1       | 1          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 17.     | X-NOR gate | <p>Boolean expression:<br/><math>Y = \overline{A \oplus B} = A \odot B</math> or <math>Y = \overline{\overline{A}B + A\overline{B}}</math></p> <p>Logic symbol:</p>   $Y = \overline{\overline{A} \cdot \overline{B} + AB}$ <table><tr><th colspan="2">Input</th><th>Output</th></tr><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> | Input             |  | Output | A | B | Y | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |                                                                                      |
| Input   |            | Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| A       | B          | Y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0       | 0          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 0       | 1          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1       | 0          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |
| 1       | 1          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |  |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                      |